

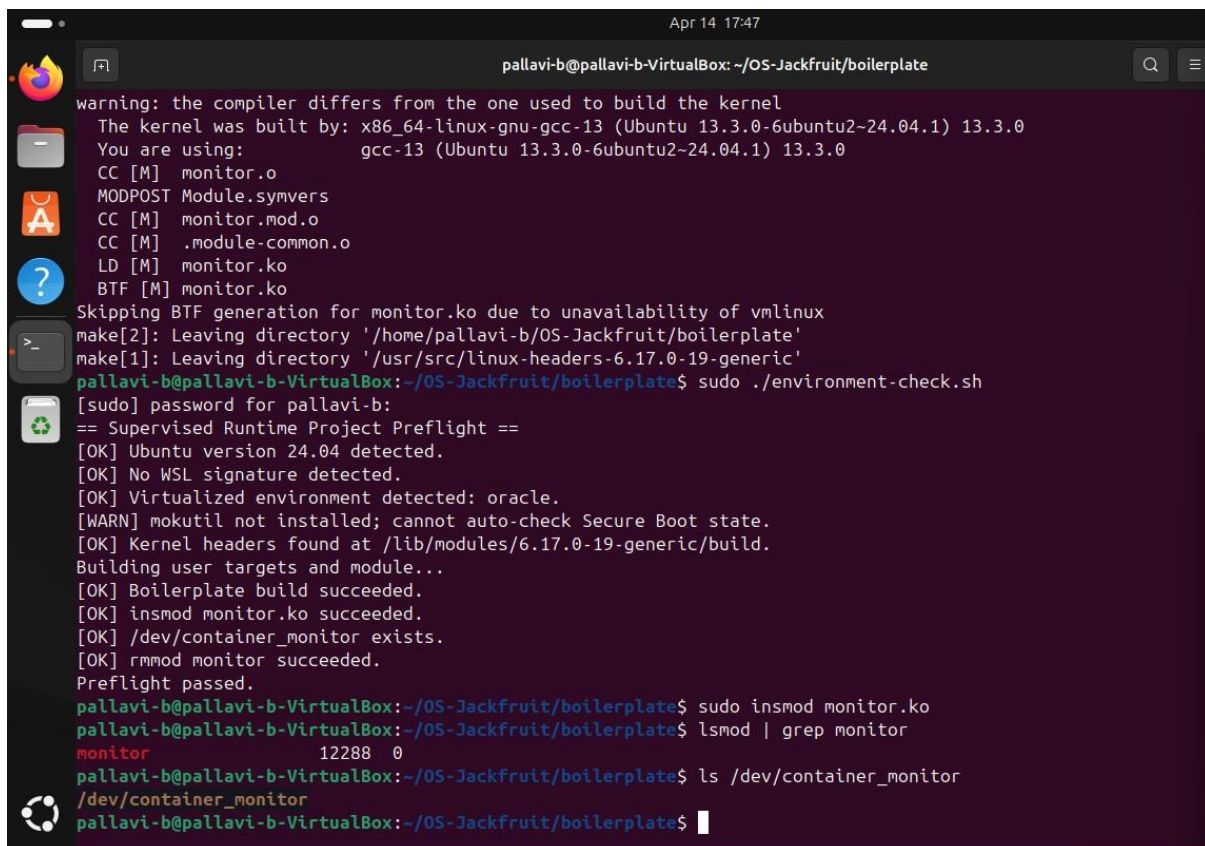
OS-JACKFRUIT

Multi-Container Runtime

Team Mem1:Pallavi B SRN:PES1UG24AM906 SEC: C

Team Mem2:Kirthana S SRN:PES1UG24AM136 SEC: C

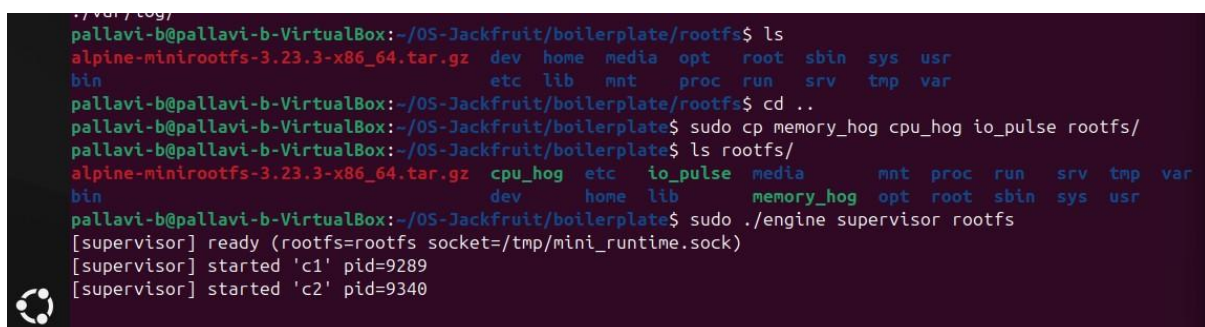
Screenshots:



```
Apr 14 17:47
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate

warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
You are using:          gcc-13 (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
CC [M] monitor.o
MODPOST Module.symvers
CC [M] monitor.mod.o
CC [M] .module-common.o
LD [M] monitor.ko
BTF [M] monitor.ko
Skipping BTF generation for monitor.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/pallavi-b/OS-Jackfruit/boilerplate'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-19-generic'
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./environment-check.sh
[sudo] password for pallavi-b:
== Supervised Runtime Project Preflight ==
[OK] Ubuntu version 24.04 detected.
[OK] No WSL signature detected.
[OK] Virtualized environment detected: oracle.
[WARN] mokutil not installed; cannot auto-check Secure Boot state.
[OK] Kernel headers found at /lib/modules/6.17.0-19-generic/build.
Building user targets and module...
[OK] Boilerplate build succeeded.
[OK] insmod monitor.ko succeeded.
[OK] /dev/container_monitor exists.
[OK] rmmod monitor succeeded.
Preflight passed.
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo insmod monitor.ko
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ lsmod | grep monitor
monitor                12288  0
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ ls /dev/container_monitor
/dev/container_monitor
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$
```

TASK 1:



```
~/OS-Jackfruit/boilerplate$ ls
alpine-minirootfs-3.23.3-x86_64.tar.gz  dev  home  media  opt  root  sbin  sys  usr
bin                                     etc  lib   mnt   proc  run  srv  tmp  var
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate/rootfs$ cd ..
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo cp memory_hog cpu_hog io_pulse rootfs/
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ ls rootfs/
alpine-minirootfs-3.23.3-x86_64.tar.gz  cpu_hog  etc  io_pulse  media      mnt  proc  run  srv  tmp  var
bin                                     dev      home  lib       memory_hog  opt  root  sbin  sys  usr
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor rootfs
[supervisor] ready (rootfs=rootfs socket=/tmp/mini_runtime.sock)
[supervisor] started 'c1' pid=9289
[supervisor] started 'c2' pid=9340
```

```
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start c1 rootfs "cpu_hog 20"
[sudo] password for pallavi-b:
ok
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start c2 rootfs "cpu_hog 20"
ok
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
c2          pid=9340    state=exited  soft= 40MiB hard= 64MiB code=127 sig=0   start=18:00:26
c1          pid=9289    state=exited  soft= 40MiB hard= 64MiB code=127 sig=0   start=17:59:40

pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start c4 rootfs "/cpu_hog 100"
ok
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
c4          pid=15111   state=running soft= 40MiB hard= 64MiB code=0    sig=0   start=18:17:11
c3          pid=11641   state=exited  soft= 40MiB hard= 64MiB code=0    sig=0   start=18:15:28
c2          pid=10204   state=exited  soft= 40MiB hard= 64MiB code=0    sig=0   start=18:08:35
c1          pid=10180   state=exited  soft= 40MiB hard= 64MiB code=0    sig=0   start=18:08:25
c2          pid=9340    state=exited  soft= 40MiB hard= 64MiB code=127 sig=0   start=18:00:26
c1
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$
```

TASK 2:

```
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
c2          pid=10204   state=exited  soft= 40MiB hard= 64MiB code=0    sig=0   start=18:08:35
c1          pid=10180   state=exited  soft= 40MiB hard= 64MiB code=0    sig=0   start=18:08:25
c2          pid=9340    state=exited  soft= 40MiB hard= 64MiB code=127 sig=0   start=18:00:26
c1          pid=9289    state=exited  soft= 40MiB hard= 64MiB code=127 sig=0   start=17:59:40

pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ ls rootfs/
alpine-minirrootfs-3.23.3-x86_64.tar.gz  cpu_hog  etc  io_pulse  media  mnt  proc  rootfs
bin                                         dev     home  lib       memory_hog  opt  root  sys

pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine logs c1
/bin/sh: cpu_hog: not found
cpu_hog alive elapsed=1 accumulator=8488015175725468995
cpu_hog alive elapsed=2 accumulator=6033449358966712512
cpu_hog alive elapsed=3 accumulator=4339841198134536119
cpu_hog alive elapsed=4 accumulator=7292795584196533636
cpu_hog alive elapsed=5 accumulator=8615908445189209954
cpu_hog alive elapsed=6 accumulator=2397641973987662503
cpu_hog alive elapsed=7 accumulator=1767593211116908633
cpu_hog alive elapsed=8 accumulator=12075446813871187839
cpu_hog alive elapsed=9 accumulator=11390401373559937977
cpu_hog alive elapsed=10 accumulator=6427755514765479242
cpu_hog alive elapsed=11 accumulator=5547866628412097498
cpu_hog alive elapsed=12 accumulator=13225349365678508617
cpu_hog alive elapsed=13 accumulator=14462013591419234240
cpu_hog done duration=20 accumulator=6801000933385675702
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine stop c1
container not found or not running
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$
```

TASK 3:

```
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run c3 rootfs "/cpu_hog 5"
[run] waiting for container to finish...
state=exited code=0 signal=0
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ cat logs/c3.log
cpu_hog alive elapsed=1 accumulator=6714212733155315941
cpu_hog alive elapsed=2 accumulator=6618240562122467652
cpu_hog alive elapsed=3 accumulator=16239123005603313596
cpu_hog alive elapsed=4 accumulator=951432212908341971
cpu_hog alive elapsed=5 accumulator=18109593245130871426
cpu_hog alive elapsed=6 accumulator=686108703922880681
cpu_hog alive elapsed=7 accumulator=3401053677491740527
cpu_hog alive elapsed=8 accumulator=12233038655658966537
cpu_hog alive elapsed=9 accumulator=2041230254743977408
cpu_hog alive elapsed=13 accumulator=7170637677605683577
cpu_hog alive elapsed=14 accumulator=16393574937059400447
cpu_hog alive elapsed=15 accumulator=11827388193847270057
cpu_hog alive elapsed=16 accumulator=14377626064443733320
cpu_hog alive elapsed=17 accumulator=13522037886052027350
cpu_hog alive elapsed=18 accumulator=7607683809324154061
cpu_hog alive elapsed=19 accumulator=12651013183571407044
cpu_hog alive elapsed=20 accumulator=2518993587427591787
cpu_hog alive elapsed=21 accumulator=4434451414469635023
cpu_hog alive elapsed=22 accumulator=13965148304984013561
```



```
Apr 14 18:21
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate
cpu_hog alive elapsed=25 accumulator=6534811994822408123
cpu_hog alive elapsed=26 accumulator=6760259344068015841
cpu_hog alive elapsed=30 accumulator=1147220970640167515
cpu_hog alive elapsed=31 accumulator=5364839943822199549
cpu_hog alive elapsed=32 accumulator=7264092236270666038
cpu_hog alive elapsed=33 accumulator=4499155384538955452
cpu_hog alive elapsed=34 accumulator=14639179011983481384
cpu_hog alive elapsed=35 accumulator=12865104147413858319
cpu_hog alive elapsed=36 accumulator=12712876961592311047
cpu_hog alive elapsed=37 accumulator=1128544247641671581
cpu_hog alive elapsed=38 accumulator=386858397204211678
cpu_hog alive elapsed=39 accumulator=6015178973589781073
cpu_hog alive elapsed=54 accumulator=15006476381075981262
cpu_hog alive elapsed=55 accumulator=1791846428048886155
cpu_hog alive elapsed=56 accumulator=632456040251411127
cpu_hog alive elapsed=57 accumulator=7384659404389094684
cpu_hog alive elapsed=58 accumulator=1096823671401906327
cpu_hog alive elapsed=59 accumulator=9557075351983169063
cpu_hog alive elapsed=60 accumulator=411175023313787733
cpu_hog alive elapsed=61 accumulator=7619184934965498454
cpu_hog alive elapsed=62 accumulator=1502981503678653814
cpu_hog alive elapsed=63 accumulator=1760525071273811290
cpu_hog alive elapsed=64 accumulator=5842404482700947076
cpu_hog alive elapsed=65 accumulator=4295718307275054603
cpu_hog alive elapsed=79 accumulator=13487021162472607042
cpu_hog alive elapsed=80 accumulator=1054438502070234777
cpu_hog alive elapsed=81 accumulator=1704060486756370775
cpu_hog alive elapsed=82 accumulator=1368444948233565960
cpu_hog alive elapsed=83 accumulator=3608759285278082025
cpu_hog alive elapsed=84 accumulator=11782575117011700673
cpu_hog alive elapsed=85 accumulator=14185679886525056786
```

TASK 4:

```
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate$ sudo ./engine run m1 rootfs "/memory_hog 8 500"
[run] waiting for container to finish...
state=killed code=0 signal=9
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate$ dmesg | tail
dmesg: read kernel buffer failed: Operation not permitted
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate$ sudo dmesg | tail
[ 4258.085869] [container_monitor] pid=16902 (container=c3) gone, removing entry
[ 4292.143385] workqueue: vmstat_update hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
[ 4380.408484] [container_monitor] Registered container=m1 pid=17838 soft=41943040 hard=67108864
[ 4382.867403] [container_monitor] SOFT LIMIT container=m1 pid=17838 rss=42409984 limit=41943040
[ 4384.915134] [container_monitor] HARD LIMIT container=m1 pid=17838 rss=75964416 limit=67108864 - SIGKILL sent
[ 4393.101766] workqueue: delayed_vfree_work hogged CPU for >10000us 7 times, consider switching to WQ_UNBOUND
[ 4434.611571] audit: type=1400 audit(1776171183.890:414): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
[ 4444.967648] audit: type=1400 audit(1776171194.251:415): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
[ 4482.619376] audit: type=1400 audit(1776171231.904:416): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
[ 4487.622521] audit: type=1400 audit(1776171236.907:417): apparmor="DENIED" operation="open" class="file" profile="snap
.firefox.firefox" name="/proc/pressure/memory" pid=4141 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=100
0 ouid=0
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate$
```

TASK 5:

```
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate$ sudo ./engine start c8 rootfs "/cpu_hog 30" --nice 10
sudo ./engine start c9 rootfs "/cpu_hog 30" --nice 10
sudo ./engine start c10 rootfs "/cpu_hog 30" --nice -5
ok
ok
ok
ok
pallavi-b@pallavi-b-VirtualBox: ~/OS-Jackfruit/boilerplate$ top
top - 18:32:59 up 1:23, 1 user, load average: 1.05, 0.98, 1.75
Tasks: 233 total, 6 running, 226 sleeping, 1 stopped, 0 zombie
%Cpu(s): 28.2 us, 25.6 sy, 35.9 ni, 0.0 id, 0.0 wa, 0.0 hi, 10.3 si, 0.0 st
MiB Mem : 3914.8 total, 384.7 free, 2233.9 used, 1395.8 buff/cache
MiB Swap: 3916.0 total, 3550.8 free, 365.2 used, 1680.9 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
 18547 root        15   -5   1012    728    676 R   94.9   0.0   0:06.67  cpu_hog
 18534 root        30   10   1012    732    676 R   71.5   0.0   0:05.29  cpu_hog
 18524 root        30   10   1012    724    676 R   24.9   0.0   0:01.42  cpu_hog
 5012 pallavi+  20   0 3523016 769368 118476 R   3.7  19.2  24:44.75  Isolated Web Co
 2623 pallavi+  20   0 3905632 169764 89108 R   2.0   4.2   6:43.44  gnome-shell
 244 root        19  -1  51460  17396  16624 S   0.8   0.4   0:15.23  systemd-journal
18554 pallavi+  20   0  14520   6036   3820 R   0.8   0.2   0:00.08  top
 6914 pallavi+  20   0  701452  54116  41236 S   0.6   1.3   0:40.32  gnome-terminal-
 15 root        20   0      0      0      0 I   0.3   0.0   0:21.30  rcu_preempt
 4141 pallavi+  20   0  11.9g 520116 201432 S   0.3  13.0  12:58.29  firefox
 4461 pallavi+  20   0 2697204 151192 101852 S   0.3   3.8   0:36.20  Privileged Cont
 5683 pallavi+  20   0 2644756 93024  66040 S   0.3   2.3   0:13.96  Isolated Servic
 9599 pallavi+  20   0 1457544 173664 108536 S   0.3   4.3   0:14.08  nautilus
10833 root        20   0      0      0      0 I   0.3   0.0   0:03.91  kworker/u9:6-events freezable pwr efficient
```

```

MiB Swap: 3916.0 total, 3550.8 free, 365.2 used. 1680.9 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
 18547 root        15   -5   1012    728    676  R   94.9    0.0   0:06.67 cpu_hog
 18534 root        30   10   1012    732    676  R   71.5    0.0   0:05.29 cpu_hog
 18524 root        30   10   1012    724    676  R   24.9    0.0   0:01.42 cpu_hog
 5012  pallavi+   20    0 3523016 769368 118476 R    3.7   19.2  24:44.75 Isolated Web Co
 2623  pallavi+   20    0 3905632 169764  89108 R    2.0    4.2   6:43.44 gnome-shell
 244  root       19   -1   51460   17396  16624 S    0.8    0.4   0:15.23 systemd-journal
18554  pallavi+   20    0  14520   6036   3820 R    0.8    0.2   0:00.08 top
 6914  pallavi+   20    0 701452   54116  41236 S    0.6    1.3   0:40.32 gnome-terminal-
 15  root       20    0      0      0      0  I    0.3    0.0   0:21.30 rcu_preempt
 4141  pallavi+   20    0  11.9g 520116 201432 S    0.3   13.0  12:58.29 firefox
 4461  pallavi+   20    0 2697204 151192 101852 S    0.3    3.8   0:36.20 Privileged Cont
 5683  pallavi+   20    0 2644756 93024  66040 S    0.3    2.3   0:13.96 Isolated Servic
 9599  pallavi+   20    0 1457544 173664 108536 S    0.3    4.3   0:14.08 nautilus
10833  root       20    0      0      0      0  I    0.3    0.0   0:03.91 kworker/u9:6-events_freezable_pwr_efficient
 1  root       20    0  23468   15016   9816 S    0.0    0.4   0:16.00 systemd
 2  root       20    0      0      0      0  S    0.0    0.0   0:00.35 kthreadd
 3  root       20    0      0      0      0  S    0.0    0.0   0:00.00 pool_workqueue_release
 4  root       20   -20      0      0      0  I    0.0    0.0   0:00.00 kworker/R-rcu_gp
 5  root       20   -20      0      0      0  I    0.0    0.0   0:00.00 kworker/R-sync_wq
 6  root       20   -20      0      0      0  I    0.0    0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
 7  root       20   -20      0      0      0  I    0.0    0.0   0:00.00 kworker/R-slub_flushwq
 8  root       20   -20      0      0      0  I    0.0    0.0   0:00.00 kworker/R-netns
12  root       20    0      0      0      0  I    0.0    0.0   0:00.00 kworker/u8:0-ipv6_addrconf

[2]+  Stopped                  top
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$

```

TASK 6:

```

./var/log/
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate/rootfs$ ls
alpine-minirootfs-3.23.3-x86_64.tar.gz  dev  home  media  opt  root  sbin  sys  usr
bin                                     etc  lib  mnt  proc  run  srv  tmp  var
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate/rootfs$ cd ..
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo cp memory_hog cpu_hog io_pulse rootfs/
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ ls rootfs/
alpine-minirootfs-3.23.3-x86_64.tar.gz  cpu_hog  etc  io_pulse  media  mnt  proc  run  srv  tmp  var
bin                                     dev  home  lib  memory_hog  opt  root  sbin  sys  usr
pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor rootfs
[supervisor] ready (rootfs=rootfs socket=/tmp/mini_runtime.sock)
[supervisor] started 'c1' pid=9289
[supervisor] started 'c2' pid=9340
[supervisor] started 'c1' pid=10180
[supervisor] started 'c2' pid=10204
[supervisor] started 'c3' pid=11641
Container 'c3' already running
Container 'c3' already running
Container 'c3' already running
[supervisor] started 'c4' pid=15111
[supervisor] started 'c3' pid=16902
[supervisor] started 'm1' pid=17838
[supervisor] started 'c4' pid=18172
[supervisor] started 'c5' pid=18216
[supervisor] started 'c8' pid=18524
[supervisor] started 'c9' pid=18534
[supervisor] started 'c10' pid=18547

```

^C[supervisor] shutting down pallavi-b@pallavi-b-VirtualBox:~/OS-Jackfruit/boilerplate\$